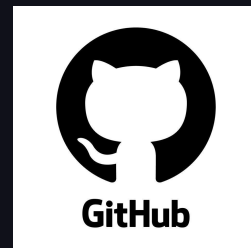




Community

Codetopia Mentorship Program - 2026

Git & GitHub



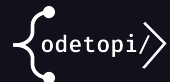
Week 2 Capstone
&
Project

Safe History Audit

Capstone Goal

Demonstrate your understanding of:

- `git restore`
- `git restore --staged`
- `Git revert`
- `Git reset --soft`
- Safe history management
- Commit hygiene
- Undoing mistakes correctly
- Documenting Git workflows





Capstone Requirements

Documentation Files:

- `undo-techniques.md`
- `restore-vs-revert.md`
- `reset-notes.md`
- `workflow-log.md`
- `reflection.md`





undo-techniques.md Must Include

Document:

- `git restore`
- `git restore --staged`
- `git revert`
- `git reset --soft`

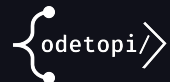




undo-techniques.md Must Include

For each command explain:

- Purpose
- When to use it
- Example command (Screenshots and should be in the screenshots folder)
- Expected result (Screenshots and should be in the screenshots folder)





restore-vs-revert.md Must Include

Create a comparison table.

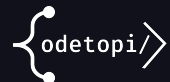
- Command, Change History, Safe for Shared Repo? and Typical Use Case

It should include:

- `git restore`
- `git restore --staged`
- `git revert`
- `git reset --soft`



Note that the table should be created in the .md file you have created. Click [Here](#) to know how to create a table in .md (markdown syntax)



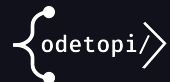
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reset-notes.md Must Include

Explain:

- What is a soft reset?
- Difference between (Use table):
 - soft reset
 - mixed reset
 - hard reset
- Include examples (Screenshots All stored in the screenshot folder)



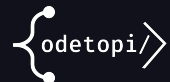


workflow-log.md Must Include

Record every operation performed

Example

1. Added bad content to recipes.md
2. Used git restore recipes.md
3. Verified changes removed
4. Created bad commit
5. Used git revert HEAD
6. New revert commit created

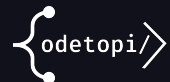




reflection.md Must Include

Answer:

- When should you use git restore?
- When should you use git restore --staged?
- Why is git revert safer than rewriting history?
- What is the difference between revert and reset?
- What mistake did you make and how did you recover?
- What did you learn during Week 2?



Capstone requirements

Expected Folder Structure:

```
codetopia-mentorship-program/  
├── README.md  
├── week-1-foundation/  
├── week-2-safe-history/  
│   ├── session-1/  
│   │   ├── undo-techniques.md  
│   │   ├── restore-vs-revert.md  
│   │   └── screenshots/  
│   │       ├── restore-command.png  
│   │       └── restore-staged.png  
│   ├── session-2/  
│   │   ├── reset-notes.md  
│   │   ├── workflow-log.md  
│   │   └── screenshots/  
│   │       ├── revert-log.png  
│   │       ├── soft-reset.png  
│   │       └── recommit.png  
│   └── capstone/  
│       ├── reflection.md  
│       └── screenshots/  
│           └── final-clean-history.png  
├── week-3-branching/  
├── week-4-collaboration/  
└── ...
```



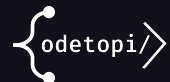
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Capstone requirements

The repository must contain:

- Week 1 content
- Week 2 content
- Proper folder structure
- Markdown documentation
- Screenshots organized correctly
- Clean commit history





Capstone requirements

Must demonstrate:

Scenario 1 - Discard Working Changes

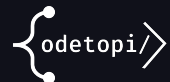
- `git restore file.md`

Take screenshots, before and after

Scenario 2 - Unstage Changes

- `git restore --staged file.md`

Take screenshots, `git status` before and `git status` after





Capstone requirements

Scenario 3 - Revert a Bad Commit

- `git revert HEAD`

Take screenshots, before revert, after revert and git log

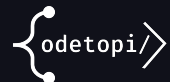
Scenario 4 - Soft Reset

- `git reset --soft HEAD~1`

Then

- `git commit -m "<better commit message>"`

Take screenshots, before reset, after reset and recommit

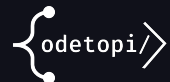


Capstone Brief



Requirement

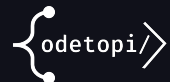
- Use the Week 1 repo as the base
- At least one commit reverted cleanly using `git revert` (history preserved)
- At least one soft reset demonstrated — changes re-committed with a better message
- `git log` shows a clean, linear history with no orphan commits





Week 2 Project

- Create a new file:
 - `week-2-safe-history/capstone/experiment.md`
- Make several commits.
- Intentionally create mistakes:
 - wrong content
 - wrong staging
 - wrong commit message
 - unwanted commit
- Recover using:
 - `git restore`
 - `git restore --staged`
 - `git revert`
 - `git reset --soft`





Week 2 Project

- Document every action in:
 - `workflow-log.md`
- Make several commits.
- Intentionally create mistakes:
 - wrong content
 - wrong staging
 - wrong commit message
 - unwanted commit
- Recover using:
 - `git restore`
 - `git restore --staged`
 - `git revert`
 - `git reset --soft`

